

A Deterministic Pipeline for Outbound Lead Generation: Confining a Language Model to Fuzzy Judgment in an Industrial Sales Workflow

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InstaLILY AI Case Study: DuPont Tedlar, Graphics & Signage

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Abstract

Outbound prospecting for a specialty materials business unit is difficult because the evidence that qualifies an account is scattered across association rosters, event exhibitor directories and technical product pages rather than concentrated in any database. Generative models are well suited to reading those sources, but systems that delegate the qualification decision itself to a model produce scores that cannot be reproduced or argued with by the sales manager who must act on them. This paper presents a lead generation and outreach agent for DuPont Tedlar's Graphics and Signage unit built on the opposite bias: a seven-stage deterministic pipeline in which scoring, deduplication, retries, persistence and provider selection are ordinary Python, and the model is confined to two fuzzy tasks: reading unstructured pages and writing prose over facts already verified. The system sourced 97 companies, enriched 90 from their own websites, qualified 16 as tier A or B, identified 32 decision-makers and produced 29 outreach drafts, at \$0.63 across 108 model calls, or \$0.039 per qualified lead. Every claim carries the URL it was read from, missing data lowers confidence rather than earning an inferred value, and 44 expected failures were recorded as data rather than terminating the run.

Introduction

DuPont Tedlar is a protective film sold into graphics and signage applications where ultraviolet exposure, weathering and warranty duration determine material choice. Identifying which fabricators, converters and print manufacturers are plausible buyers is a research task before it is a sales task: the qualifying signal lives in a company's own account of its durability claims and application areas, and in whether it exhibits alongside Tedlar's audience at events such as ISA Sign Expo, FESPA and PRINTING United. Assembling that list by hand is slow and inconsistent, which makes the workflow a natural candidate for automation.

The obvious strategy, and the one the brief invites, is a swarm of cooperating model-driven agents that discover, evaluate and decide. This work deliberately rejects that design. A model that assigns the score cannot explain it in terms a sales manager can dispute, gives different answers on identical inputs, and offers no place to attach a defect when a lead is wrong. The motivation is the separation of mechanism from judgment: decisions expressible as a formula are expressed as a formula, and the model is invoked only where the input is genuinely unstructured. The contributions include a pipeline whose qualification logic is deterministic and expressed as data in one configuration file, a scoring model emitting a per-component breakdown with the source URL behind each component, guardrails that reject model output citing evidence the pipeline did not supply, and an error-isolation primitive that degrades per record rather than per stage. Section 2 describes the agent workflow, Section 3 the data processing steps, and Section 4 the implementation results.

AI Agent Workflow

The pipeline runs seven stages, presented in the interface as six named agents: event and association discovery, company sourcing, deduplication and normalization, enrichment, qualification, contact discovery and outreach drafting. Orchestration is deterministic; each agent is a stage with declared inputs, a persistence boundary and its own failure isolation.

Qualification assigns a 0 to 100 score across five weighted components: industry fit (30 points), application fit (25), company size (15), event engagement (15) and pain-point alignment (15). Accounts scoring at least 75 are tier A, at least 60 tier B, and at least 40 tier C; below that they fall outside the ICP. Vocabulary, weights and thresholds are data in a single file, so retargeting at a different business unit is a configuration edit rather than a change to any stage. The model never produces the number. It receives the finished breakdown and the evidence list and writes the rationale, which is a presentation task over a decided result. Confidence is computed separately and measures how much of the score is URL-backed, penalized for a missing website or unknown size.

Contact discovery is a provider chain behind one interface: a public-web provider that follows a site's own navigation to leadership pages, a deterministic mock provider whose records always render at zero confidence with an explicit badge,

and credential-gated Clay and Sales Navigator providers implemented against their real call shapes. Candidates are ranked by functional fit rather than seniority alone, since seniority ordering places a finance director above the product manager who actually specifies materials. Outreach drafting requires the model to return both its hook fact and the URL for that fact; if the URL is not one the pipeline supplied, the draft is rejected.

Data Processing

Companies arrive through three channels that fail in different ways: exhibitor directories, ICP-shaped search queries, and a curated roster of category leaders. The failure of any one degrades recall rather than emptying the corpus. Records are deduplicated on registrable domain first and fuzzy legal name second. Every HTTP response is written to a content-addressed cache keyed by URL hash, with a sidecar recording status and timestamp; this is what makes a cached replay reproducible, and keyed search providers were routed through the same cache after it was found that they bypassed it and silently re-searched live.

Enrichment fetches each company's own website and follows its navigation to product and technical pages, because the durability, weathering and warranty language carrying the qualifying signal rarely appears on a homepage. Extracted text is retained at a deliberately generous cap: the keyword scan driving the score is nearly free, while the model sees only the leading portion, so truncating early discards evidence already paid for. The extraction prompt is null-preferring and forbidden from inferring, and a monetary figure is accepted as revenue only when a revenue word appears within sixty characters of it, which prevents a phrase such as a multi-billion-dollar market from being recorded as company turnover. All model output is parsed into a schema before reaching the database, with one repair round-trip permitted before the record is degraded. Per-record work is wrapped in an error-isolation primitive that converts expected failures into stored error rows and lets the run continue; an uncaught exception is treated as a defect rather than an outcome.

Implementation Results

A full run with search, enrichment and model providers configured sourced 97 companies and enriched 90 from their own sites, qualifying 16 as tier A or B, six tier A and ten tier B, across 10 discovered events and associations. Contact discovery returned 32 named decision-makers, and outreach produced 29 drafts, of which 28 were model-written and one fell back to a labelled template. The run consumed 108 model calls at an estimated \$0.63, or \$0.039 per qualified lead, and recorded 44 handled failures. The test suite comprises 165 hermetic tests that blank every credential and perform no network access. Results are exposed through a dashboard with score breakdowns, clickable evidence, contact cards, an outreach editor and an error tab.

The most useful evaluation was adversarial rather than aggregate: testing against the single example account named in the brief. That exercise surfaced three defects inspection had missed. The pain-point vocabulary matched too literally, failing on a company that discusses warranty extensively without using the exact phrase sought. Enrichment consumed its firmographics quota in alphabetical order, so accounts late in the corpus were never enriched. A name guard rejected precisely the pages that answered the qualification question. Correcting these moved the example account from 48 to 73 and raised corpus-wide qualified leads from 6 to 16. Relatedly, at a lower text-retention cap the median company was truncated exactly at the boundary and pain-point alignment read zero on 96 of 116 records.

Two limitations are visible in the results themselves. Live runs are not reproducible because search ranking drifts; the example account has scored 48, 61 and 73 across executions, which is why demonstration uses a cached replay. Second, none of the 32 contacts carries a verified email address, as the available people-search tier does not expose them. The interface states this rather than constructing a plausible address, the same principle that governs unknown revenue.

Conclusion

This work presents an outbound lead generation system whose qualification logic is deterministic, inspectable and cheap, and in which a language model contributes only where the input is genuinely unstructured. The resulting scores are reproducible and decomposable into components with source URLs, which makes them arguable in the way a sales manager needs. The provenance discipline is the substance of the contribution rather than a wrapper around it: null-preferring extraction, evidence-bound rationales, rejected drafts citing unsupplied URLs, and labelled placeholder records.

Several constraints bound the system. Exhibitor directories served by a common event platform are JavaScript-gated and refuse server-side clients; this is recorded as a miss rather than circumvented, since defeating anti-bot measures is out of scope and a licensed source is the correct remedy. Without a paid search key, recall on category leaders is thin, which is

why the curated roster channel exists. Because most private companies do not publish revenue, the size component is frequently zero. The evaluation covers one business unit and one corpus, so generalization remains untested. Future work centers on temporal signals, detecting why an account is worth contacting now rather than only whether it fits, and on weighting components against closed-won outcomes instead of by hand. Both are designed and deliberately unbuilt, as each requires a longitudinal record the system has only begun to accumulate.

References

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Supplementary Materials

GitHub Repository: <https://github.com/mohsin-khawaja/instalily-case-study>

Live Dashboard: <https://instalily-case-study-seven.vercel.app/>

Presentation (process and prototype): docs/Tedlar-Lead-Agent.pptx

Extended write-up: docs/Tedlar-Lead-Agent-Writeup.pdf